Biomedical Equipment & Maintenance Management System

System User Manual

Welcome to the custom Bio-Med Device Management System deployed at A.G. Padmavati's Hospital Ltd. This guide will walk you through how to use the software based on your assigned role. Please find your role below and follow the step-by-step instructions.

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1. General User Guide (Department Staff)

As a General User, you can view your department's equipment, report issues, and track the status of your requests.

How to View Your Department's Equipment

1. Navigate to the **Total Equipments** module from your dashboard.
2. View the complete list of all equipment currently assigned to your specific department.

How to Report Broken Equipment (Breakdown Intimation)

1. Click on **Breakdown Intimation** in the main menu.
2. Enter the **Asset Number** of the faulty equipment.
3. Select the **Notice Time and Date** when the issue occurred or was noticed.
4. Type a description of the problem in the **Issue** text box.
5. Upload a photo of the issue (Optional).
6. Click the **Submit** button to send the request directly to the Biomedical team.

How to Check Your Request Status (Breakdown History)

1. Go to **User Breakdown History**.
2. View the list of all the requests you have submitted.
3. Look at the **Status** column to see if your request is *Pending* or *Accepted*.
4. Read the **Action Taken** notes provided by the Biomedical staff to see what is being done to fix the issue.

How to Track Equipment Location (In or Out)

1. Navigate to the **In or Out** section.
2. View your department's inventory to see if an item is currently in your department ("In") or temporarily given to another department ("Out").
3. Click on the item to view the complete history of its movement.

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2. Biomedical Staff Guide

As Biomedical Staff, you are responsible for managing inventory, handling maintenance requests, and tracking equipment lifecycles.

Handling AMC Alerts

1. Log into the system and locate the **AMC** module.
2. Click on it to open a popup displaying all equipment with Annual Maintenance Contracts (AMC) that are due to expire within the next three months.

How to Manage Inventory (Biomedical Equipment)

1. **View & Search:** Go to the **Biomedical Equipment** module to view full stock details. Use the search box and filter options to find specific items.
2. **Export:** Set your desired filters, then click **Export to Excel** to download the stock list.
3. **Add New:** Click **Add New Asset** to register a newly arrived piece of equipment into the system.
4. **View Details & Condemn:** Click on any item in the list to view its full details. If an item is beyond repair, click the option to move it to **Condemnations**.
5. **Manage Condemnations:** Navigate to the **Condemnation** view to see all discarded items and their attached comments. To restore an item to active stock, click **Retrieve** and enter the required reason.

How to Handle Service Requests (New / Active / Received)

1. Go to **New/Active Requests** to view incoming breakdown intimations from department users.
2. Click **Accept** on a request to acknowledge it.
3. Enter updates into the **Action Taken** field (Note: General Users can see this text).
4. Update the item's status by selecting one of the following options: *Send to Company*, *Item Movement*, *Condemn*, or *Solved*.

How to Track Item Movement

1. Go to the **Item Movement Module**.
2. View which items are currently located in other departments and check their status to see if they have been returned.
3. View the chronological history of all movement for any specific piece of equipment, including its current "In/Out" status.

How to Review Work History & Solved Requests

1. Go to **Work History / Solved Requests** to see all completed maintenance.
2. Use the date filters (**From Date** to **To Date**) to narrow down the list, and click **Export to Excel** to generate a report.

3. Click on any specific equipment name to see the entire maintenance history for that single machine.
4. Review the equipment record screen displaying: *Current Status, Reported By, Reported On, Problem Date, Solved On,* and *Issue Description*.

How to Log and Schedule Maintenance

1. **Individual Update:** Click on an item and log its status as *Working*, *Not Working*, or *Company Service*. You must also enter the date, time, and your comments.
2. **View Calendar:** Click on an item to open the Maintenance Calendar to see a color-coded overview of equipment status.
3. **Bulk Update:** To send multiple items to company service at once, click **Company Maintenance** at the top of the screen. Use the search box to filter the list, check the boxes on the left side of the relevant items, and click to apply the bulk update.

Maintenance Calendar Color Codes

Color	Status
Green	Working
Red	Not Working
Blue	Company Service

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3. Sudo-Admin Guide

As a Sudo-Admin, you have total control over the system's users, roles, and permanent data records.

How to Permanently Delete Equipment

1. Navigate to the **Biomedical Equipment Stocks** module.
2. Click on the specific item you wish to remove.
3. Click the **Delete Item Permanently** button to completely remove it from the system database.

How to Manage User Roles

1. Go to the **User Management** module.
2. Click **Add Role** to create a new role.
3. Select the checkboxes next to the specific modules (e.g., Work History, Breakdown Intimation) that this role is allowed to access, and save.

How to Manage Users

1. **View Users:** Go to the **User Management** module to see a list of all staff, their assigned roles, module access, and department access.
2. **Add User:** Click **Add User**. Fill in the username and password, choose their predefined **Role** from the dropdown, and select which **Department** they have access to.
3. **Edit/Delete:** Click the **Edit** button next to a user's name to change their details or access. Click **Delete** to permanently remove their account from the system.

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4. Technical Architecture

The Bio-Med Device Management System is built upon a robust, scalable, and secure **MEAN Stack** architecture, ensuring high performance and real-time data processing across all hospital departments.

Frontend (Client-Side)

- **Framework:** Angular
- **Implementation:** Delivers a responsive, single-page application (SPA) experience. It utilizes Angular's modular component-based architecture to separate views for General Users, Biomedical Staff, and Sudo-Admins.
- **Features:** Reactive forms for accurate breakdown intimations, dynamic data binding for real-time maintenance calendar updates, and route guards to enforce access control on the client side.

Backend (Server-Side)

- **Environment & Framework:** Node.js with Express.js
- **Implementation:** Acts as the central nervous system of the application, exposing a RESTful API. It handles all business logic, including AMC expiration calculations, equipment movement tracking, and bulk maintenance updates.
- **Features:** Express middleware is utilized for request validation, error handling, and routing API calls efficiently between the client and the database.

Database Layer

- **Database:** MongoDB
- **Implementation:** A NoSQL document-based database chosen for its flexibility in handling complex, hierarchical data structures associated with medical equipment.
- **Features:** Utilizes Mongoose ODM (Object Data Modeling) to enforce strict schemas for Equipment Records, User Profiles, and Maintenance Histories, ensuring data integrity while allowing for rapid querying and reporting.

Security & Authentication

- **Protocol:** JSON Web Tokens (JWT) & Role-Based Access Control (RBAC)

- **Implementation:** The system enforces strict RBAC, completely isolating Sudo-Admin privileges from standard operational roles. Passwords are cryptographically hashed before being stored in the database.
- **Features:** API endpoints are secured using JWT authentication, ensuring that only users with the verified correct roles (e.g., Biomedical Staff vs. Department User) can perform specific CRUD (Create, Read, Update, Delete) operations.

Deployment & Infrastructure

- **Containerization:** Docker
- **Implementation:** The application components (Frontend, Backend, Database) are containerized using Docker.
- **Features:** This ensures a consistent, isolated, and secure deployment environment across the hospital's infrastructure, facilitating easier maintenance, seamless updates, and reliable performance without environmental conflicts.